

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA0301	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	T. Bhanu Teja	Reg. No.:	192472259
Section / Batch:	2024	Date:	09-08-26

Code of Conduct

I T. Bhanu Teja (192472259) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words "talked", "talking", and "talks". Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.

2. Given the input words: "relational", "conditional", "happiness", "running", design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.

3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as "optimize", "optimization", "optimized", and "optimizing". The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.

4. A large-scale information retrieval system processes input words: "compute", "computer", "computing", and "computation" to enhance search accuracy.

Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.

Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.

Ensure consistent root extraction across all variations for efficient indexing.

Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.

5. A text mining system processes input words: "decide", "decision", "decisive", and "deciding" to support semantic clustering and analysis.

Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.

Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset. Derive the morphological structure, state transitions, and final stem for each input word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

09-08-26

T. Bhanu Teja

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DSA0201

Natural Language Processing.

CO2-AT3

Q1. Morphological Processing for "analyzing", "analysis", and "analytical".

Aim:- To design a morphological processing approach that decomposes the given words into root and affixes, classifies them as inflectional or derivational, and normalization them as a common base form for efficient document indexing.

Morphological Decomposition;

word	Root	Affix	TYPE
analyzing	analyze	-ing	Inflectional
analysis	analyze	-sis	Derivational
analytical	analyze	-ical	Derivational.

Normalization: All three words are converted to the

Common base form:

analyzing
 analysis
 analytical

} → analyze

Final output
word:

analyzing
 analysis
 analytical

Morphological Structure

analyze +ing
 analyze +sis
 analyze +tical

Normalized

analyze
 analyze
 analyze

Conclusion The morphological processor identifies the root word "analyze", distinguishes inflectional and derivational forms, and normalizes all variants into a single representation. This improves document indexing, reduces duplicate entries, and increases search accuracy.

Q2: Morphological Parsing for "disagree", "agreement", and "agreeable".

Aim, to design a morphological parsing strategy that identifies prefixes, suffixes, and base forms while preserving semantic meaning for sentiment analysis.

Morphological Decomposition:

<u>word</u>	<u>prefix</u>	<u>Root</u>	<u>suffix</u>	<u>Type</u>
disagree	dis -	agree	-	Derivation
agreement	-	agree	-ment	Derivation
agreeable	-	agree	-able	Derivational

Morphological Parsing:

disagree → dis + agree
 agreement → agree + ment
 agreeable → agree + table

Normalized Root:

disagree
 agreement
 agreeable

} → agree

Conclusion - The Parser successfully identifies the common root "agra" and analyses the effect or prefix, and suffixes. This helps sentiment analysis systems understand related words while preserving their semantic differences.

Q3. Morphology - Based normalization for "govern", "government", and "governance".
Aim - To normalize related words into a common base form for topic modeling and document clustering.

Decomposition

<u>word</u>	<u>Root</u>	<u>Suffix</u>	<u>Type</u>
Govern	Govern	-	Base word
Government	Govern	-ment	Derivational
Governance	Govern	-ance	Derivational

Normalization

Govern	} → Govern.
Government	
Governance	

Final output

<u>word</u>	<u>morphological structure</u>	<u>normalize</u>
Govern	Govern	govern
Government	Govern + ment	govern
Governance	Govern + tance	govern.

Conclusion, The normalizes process maps all words forms to the common root "govern", improving document clustering and retrieval by treating words as one concept.

Q4. Morphological Analysis for "activate", "activation", and "reactivation".

Aim: To analyze derivational layers, identify prefixes and suffixes, and normalize all forms to a common root.

Decomposition

<u>word</u>	<u>prefix</u>	<u>Root</u>	<u>Suffix</u>	<u>Type</u>
activate	—	active	-ate	Derivational
activation	—	activate	-ion	Derivational
reactivation	re-	activate	-ion	Derivational.

Classification

* activate is formed by adding -ate to create a verb.

* Activation adds -ion, changing the verb into a noun.

* reactivation adds the prefix re-, meaning "again", and the suffix -ion.

All three are derivational because they change the grammatical category or meaning.

Morphological Parsing;

activate $\rightarrow$ activate + ate
activation $\rightarrow$ activate + ion
reactivation $\rightarrow$ re + activate + ion

Normalized Root:

activate
activation
reactivation
} $\rightarrow$ activate

Conclusion: The morphological analyzer extracts the common root "activate" while identifying derivational layers and semantic changes, ensuring consistent indexing & retrieval.

⑤ Morphology - Based Normalization for "create", "creates", and "creating".
Aim: TO identify inflectional suffixes, extract the common root, and normalize all grammatical forms for document retrieval.

Morphological Decomposition:-

<u>Word</u>	<u>Root</u>	<u>Suffix</u>	<u>Type</u>
create	create	-	Bare word
creates	create	-s	Inflectional
creating	create	-ing	Inflectional

Normalization

create
creates
creating

} → create

final output

word

morphological
structure

normalized form

Create

create

create

Creates

create +s

create

Creating

create +ing

create

Conclusion

The normalization process removes inflectional suffixes and converts all word forms to the common root "create". This improves indexing, search efficiency, and retrieval accuracy by treating all grammatical variants as a single term.